Re: Response to Openness to International Trade Subindex Evaluation

Dear Professor Cosar,

Thank you for your thorough and thoughtful evaluation of the Openness to International Trade subindex. Your critiques are valuable, and we appreciate the care you took in reviewing this element of our Index. We have made substantive changes and additions to this subindex based on your recommendations and hope they sufficiently address your concerns. Where we made no changes, we provide a detailed explanation. Below are our responses to your feedback and recommendations.

I. Subindex Title and Framing

> *"In terms of labeling, including free flow of capital in a subindex titled 'Openness to International Trade' is problematic. International trade implies exchange of goods and services. Free flow of capital is neither necessary nor sufficient. A country can be open to one and closed to the other and vice versa. I understand that you want to incorporate financial openness but don't want to create a ninth subindex. That makes sense. Then why not call this subindex "openness to international flows", in which case you can incorporate openness to immigration/travel & visa restrictions as well. (I don't know if the 'Labor Migration' component in Openness of Labor Market is about international migration; if it is, it could be moved here)"*

Your comments on the original subindex title, "Openness to International Trade," are valid and provide us with actionable corrections. We re-titled the subindex "Free Flow of Goods and Capital" to better represent our measurement of a market's ability to expand and circulate goods and services, as well as financial and human capital. Per your suggestion, we moved the World Bank Net Migration dataset from our Labor Market Openness subindex to this subindex to capture openness to economic migration. We do caveat that the net migration dataset likely captures more than just economically-driven migration and is not necessarily representative of the true policy openness of a country. However, it provides a useful proxy and indicator of migration flows. We address this lack of precision with a lower reliability score (4.25/5).

II. Dataset Concerns

Trade Participation

> *"The inclusion of 'Trade Participation' is also problematic since it is an outcome rather than a driver. Small countries have systematically higher trade participation. If you plot trade/GDP ratio against total GDP, it is a very strong negative relationship. So, this component makes large countries seem closed to international trade, whereas this is a result of them 'trading more with themselves' rather than policy."*

Your identification of the limitations inherent in outcome data such as trade participation is well taken. It is true that trade participation data by itself biases against nations that have larger internal markets that facilitate domestic trade. Larger economies tend to exhibit lower trade/GDP ratios because extensive internal markets substitute for cross-border exchange, while smaller countries rely more heavily on international markets to achieve scale. This relationship is well-documented.[1]

[1] Alberto Alesina, Enrico Spolaore, and Romain Wacziarg, "Economic Integration and Political Disintegration,"

However, country size does not solely determine trade participation. Governments shape trade intensity through tariffs, non-tariff barriers, and infrastructure provisions. Additionally, factors like resource endowment, physical location, and logistical advancements also determine the level of trade participation. Countries of similar size can have varied trade/GDP ratios driven by institutional and policy design rather than geography alone.[2]

Excluding the metric of trade participation would risk discounting smaller nations that maintain higher trade participation relative to their size through institutional commitments, investments, and policy. As a result, we present trade participation as one component element of the aggregate subindex and utilize the synthesis and weighting of multiple datasets to mitigate bias inherent in any single indicator. We believe that this complementary approach presents a more comprehensive reflection of the multidimensional nature of international exchanges of goods and services.

OECD STRI

> *"OECD STRI is a good one, but only measures Non-tariff barriers in services. Note that some of its sub-indices are actually restrictions for indirect investment. International Trade and Services has four modes. Mode 3 is effectively FDI. Since you are already using OECD's FDI Restrictions Index, it would make sense to use only Modes 1-2 STRI scores from OECD rather than the aggregate STRI index itself. Also, UNCTAD has good data on non-tariff measures for trade in goods."*

We are glad you agree that the OECD STRI is a good measure. The OECD database does not publish data that delineates between the modes you mention, however, it does provide data values broken out by industry sector and at the aggregate level per country. We utilize the aggregate index value per country in this subindex published by the OECD, which breaks the index out by five policy areas: restrictions on foreign entry, restrictions to movement of people, barriers to competition, regulatory transparency, and other discriminatory measures.[3]

Although very detailed and published by a reliable source, the UNCTAD TRAINS database only presents bilateral or product-specific data on trade barriers that would be difficult to combine into an aggregate index score per country-year instance.

Hamilton Indicator

> *"I find the Hamilton Indicator problematic. A low mean level with high variance could be driven by political economy forces in an otherwise open economy. Take the case of the US: pre-2017, the US was an open economy with some sectors enjoying persistently high protection. These are typically sectors with concentrated geographical concentration and organized lobbying. The sugar lobby being case in point. In sum, a low average level of tariffs with high variance does not necessarily indicate a deliberate industrial policy. It could just be a result of political economy and lobbying."*

Your concern about the potentially distortive effects of organized lobbying and other political economy factors on the Hamilton indicator is valid and helpful. To account for those potential

American Economic Review 90, no. 5 (2000): 1276–1296; Jeffrey A. Frankel and David Romer, "Does Trade Cause Growth?" *American Economic Review* 89, no. 3 (1999): 379–399.

[2] Arvind Subramanian and Shang-Jin Wei, "The WTO Promotes Trade, Strongly but Unevenly," *Journal of International Economics* 72, no. 1 (2007): 151–175; Douglas A. Irwin, *Free Trade under Fire*, 5th ed. (Princeton, NJ: Princeton University Press, 2020).

[3] https://www.oecd.org/en/topics/sub-issues/services-trade-restrictiveness-index.html

biases, we now weight the combined tariff level and variance datasets by V-DEM's Particularistic vs Public Goods indicator (v2dlencmps) to proxy whether tariff policy reflects rent-seeking or political motivation rather than systematic policy design. The inclusion of this measure allows us to adjust for institutional variation in policymaking incentives. Though this adjustment is not explicitly targeted at trade policies, it improves our ability to distinguish between targeted and particularistic policy.

Herfindahl-Hirschman Index

> *"Similarly, the concentration of a country's trade across trade partners captured by the Herfindahl-Hirschman Index is to some extent driven by geography and free trade agreements (FTA). An FTA is a liberalizing act, but it also tends to increase the concentration index by diverting trade."*

Your comments on the use of the HHI index and the potentially concentrating effects of FTAs are very useful. We recognize that the HHI may misrepresent the effects of liberalizing policies (like FTAs) and partially reflect geographic proximity to preferred partners. However, our intention is to capture the breadth and diversification of a country's integration into global markets. The HHI measures whether economic actors in a given economy are embedded within a wide international marketplace or are dependent on internal or a narrow set of bilateral trade relationships. Even though FTAs are liberalizing acts, a market concentrated in a small number of external trade partners is more vulnerable to political pressure and economic coercion. Broader diversification provides autonomy and multiple channels for exchange and competition.

Though the HHI evaluates structural outcomes rather than the intent of policy, it is an important measure of economic integration and the degree to which markets can operate within a competitive global system. We welcome any further thoughts on whether trade partner concentration should be interpreted in this manner, and view this as an important area for continued theoretical and empirical refinement within this subindex.

III. Additional Dataset Updates

FDI Switch

Though your review did not note concern with the FDI Net Inflows dataset, your suggested reframing from "Openness to International Trade" to "Free Flow of Goods and Capital" motivated a re-adjustment to our FDI data. To better capture the new "Flow of Goods and Capital" subindex, we switched FDI flows from Net Inflows to Total Flows (absolute value of Inflows + absolute value of Outflows). We believe that this provides better information on the ability of capital to flow cross-borders. Your comments helped us see that using net FDI inflows conflates openness with a country's savings–investment balance, thus potentially misclassifying net capital exporters as less integrated. By shifting to total FDI flows, the measure more accurately reflects the intensity of cross-border investment linkages and provides a more valid proxy for overall capital mobility.

Economic Complexity Index

Additionally, we added the Harvard Growth Lab's Economic Complexity Index (ECI) to the Free Flow of Goods component to capture the structural capability of economies to sustain participation in diverse markets. The ECI measures the accumulated productive knowledge embedded across firms, workers, and institutions that allows an economy to produce a wide variety of sophisticated goods. This is a critical addition because the free flow of goods is not solely determined by tariff policy or trade volume, but also by whether an economy possesses the underlying productive structure necessary to generate output in diverse markets. A more complex economy reduces dependence on narrow sectors and enables participation across multiple export categories and global value chains. As a result, we use the ECI to proxy the depth and adaptability of a market. Although this is not a direct measure of trade policy, we believe that it represents the institutional commitments necessary for sophisticated production, and distinguishes economies that are formally open but structurally constrained from those capable of maintaining broad and diverse flows of goods over time.

Concluding Remarks

Your review raised three concerns with the original subindex: 1) that the prior "Openness to International Trade" framing did not represent our indicators; 2) that trade participation risked capturing outcomes that did not reflect the orientation of a country's trade policy; and 3) that the Hamilton Indicator could reflect political economy dynamics rather than policy orientation.

In response, we substantially revised both the conceptual framing and construction of the subindex. We renamed the subindex "Free Flow of Goods and Capital" to more precisely reflect its scope and created three components that focus on goods, financial capital, and human capital. We added net migration data to better capture cross-border human capital flows. We also replaced net FDI inflows with total FDI flows to measure the overall volume of financial capital movement, and incorporated the Economic Complexity Index to account for the structural productive capabilities that enable sustained trade participation.

We further addressed concerns regarding political economy distortions embedded in the Hamilton Indicator by weighting tariff level and variance data with V-Dem's indicator of particularistic versus public goods provision, allowing the indicator to better distinguish between systematic trade policy and sector-specific rent-seeking. Finally, we retain the HHI as a structural indicator of whether economies can integrate into international markets rather than bilateral or narrow ones.

Together, these revisions shift the subindex toward a multidimensional framework that captures both institutional openness and the underlying capacity required to sustain broad, multidirectional flows of goods and capital. We would be happy to discuss any of these points further, and we hope to continue this productive dialogue about how to strengthen the Flow of Goods and Capital subindex.

Sincerely,

UVA Democracy and Capitalism Lab